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Research Program/Project/Study Title: Proximate and Microbial Analysis of MARC & Blue Ginger Souring Mix	
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(7) Sector/Commodity Agriculture/Fruit and Vegetables	(8) Discipline/Field of Specialization Food Development
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(9)Abstract: The study focused on the formulation and evaluation of sensory qualities and acceptability of mangrove apple and root crop (gabi, taro, cassava) powders. The evaluation covered aspects such as appearance and texture of the powders, and appearance, aroma, and taste when applied in soup. Significant differences in sensory qualities and acceptability were assessed using a completely randomized design (CRD). Three different treatments of the product underwent two trials: one evaluated by a panel of experts, and the second by 100 consumers for preference testing. Data were collected using score cards with a nine-point hedonic scale, and analyzed using mean scores. In terms of general acceptability, including appearance, aroma, and taste, Treatment B received the highest mean score with a qualitative description of “liked extremely.” Treatment A followed closely with a similar qualitative description, while Treatment C had the lowest mean score, described as “liked very much.” There was no significant difference among the three products in terms of texture.

Keywords: Mangrove Apple, Root Crops, Gabi, Taro, Cassava, Langkawas

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(10) Introduction

Souring agents and seasonings play a crucial role in enhancing the flavor of food, bringing out its inherent taste and improving texture. Seasoning, which involves using herbs, salt, and pepper, acts as a flavor foundation that stimulates appetite and enhances the appearance of meals. It helps balance intense flavors such as sweet, bitter, or sour. Seasonings can be derived from various plant parts, including leaves, fruits, and stems. Dried fruits like lime, lemon, mangoes, and tamarind are commonly used to prevent spoilage and extend shelf life. This study focuses on utilizing locally sourced fruits from the Philippines, particularly the mangrove apple (*Sonneratia ovata*), to create a nutritious, preservative-free powdered mix. The mangrove apple, also known as "pirara," is rich in nutrients and often underutilized. Processing this fruit into a powdered mix, combined with root crops (gabi, taro, cassava) and blue ginger (*Alpinia galanga*), offers a sustainable and culturally significant food source.

Mangrove apples grow in coastal areas and are known for their flavor and medicinal properties. They are high in vitamin C and contain beneficial phenolic compounds with antioxidant properties (Ray et al., 2015). Despite their nutritional value, these fruits are often overlooked and left to waste. By transforming mangrove apples into a powdered mix, this study aims to create a value-added product that can provide additional income for coastal communities and offer a wholesome food option free from preservatives. These plants are valuable not only for coastal protection but also for their nutritional content. In Capiz, the mangrove apple is common in swampy areas near fish ponds, yet many locals are unaware of its edibility and nutritional benefits. By processing mangrove apples into powdered mixes, this study hopes to elevate the fruit's value and utility.

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The study aims to inform product development and consumer preferences, promoting sustainable and innovative food choices. By incorporating community feedback through focus groups, questionnaires, and sensory evaluations, the research seeks to develop a product that resonates with local tastes and cultural preferences, fostering a sense of community ownership and pride.

Ultimately, this research addresses the need for wholesome, locally sourced food options, especially in areas with limited access to diverse products. The goal is to determine whether the formulation and acceptability of the mangrove apple-root crop-blue ginger powdered mix can meet nutritional needs and provide a viable solution for enhancing food security and health.

This research also aims to clarify the methods used to achieve the best stability and quality in powdered mixes, addressing the scientific complexities involved in their formulation. The unique combination of mangrove apple, root crops, and blue ginger provides a delicious and healthful meal option deeply rooted in local culture. Gathering community opinions and preferences is essential for ensuring the product's successful integration into local diets. This study seeks to address the nutritional needs of communities by providing a sustainable, locally sourced food option.

Statement of the problem:

1. This study was conducted to find out if the Mangrove apple (*Sonneratia ovata*) with three (3) varieties of root crops (gabi, taro, and cassava) with blue ginger can be utilized in making souring seasoning mix. Furthermore, it sought to; Describe the sensory qualities of a mangrove apple-root crop with blue ginger as a powder mix in terms of appearance and texture and when applied in soup in terms of appearance, aroma, and taste. Determine the general acceptability of mangrove apple-root crop and blue ginger as a souring mix in terms of appearance and texture and when applied in soup in terms of appearance, aroma, and taste. Determine the pH level of the mangrove apple root-crop with blue ginger powder mix when applied in soup; and Submit the best treatment of the formulation and acceptability of mangrove apple-root crops with Blue Ginger powder mix for microbial analysis and proximate analysis.

Related Studies:

Recent studies show mangrove fruits have the potential to be an essential ingredient in processed foods that are useful for the growth, development, and intelligence of humans. Many people, especially coastal communities on Java Island, have used mangrove fruits as raw material in processed foods, such as syrups and other types of food (Jariyah et al., Citation2014). In spite of these roles, only some studies realized the potential uses of mangrove fruits as a food and nutritional source for the coastal community in the tropical region of this part of the world. With

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the increasing human population alongside depleting natural resources, there is a need to explore the nutritive potential of mangroves to extend their possible uses further. Previous studies

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conducted in Southeast Sulawesi have quantified mangrove biomass and carbon stocks, the potential uses of mangrove leave as a green tea or other medicinal values, and the role mangroves play as a biofilter of heavy metal pollutants (Analuddin et al., 2016; Septiana et al., 2016).

Carugati et al. (2018) studied mangroves as being among the world's most prolific maritime ecosystems, providing a unique habitat opportunity for many species as well as essential commodities and services for humans, with significant biomass and economic values. Many terrestrial and marine creatures, including many commercial species and juvenile reef fish, rely on these forests at the land-sea interface for food, breeding grounds, and nursery sites. Mangroves are also crucial for human survival and livelihoods, as they are heavily used for food, timber, fuel, and medicine. They protect against disasters, including tsunamis, tropical cyclones, and tidal bores, as well as reducing shoreline erosion.

The ripe mangrove apple is a potential source of natural antioxidants owing to its significant antioxidant activities. Phytochemical analysis of fruit extract revealed the presence of carbohydrate, protein, flavonoid, and phenolic compounds. It can enhance the capability of the antioxidant human body and, therefore, reduce the risk of chronic diseases. It has a sweet taste and is usually consumed fresh or made into juice. A limited number of studies have mentioned the processing of this functional fruit.

Spices

Spices are the dried, pleasantly aromatic parts of the plants. More specifically, as defined by the Food and Drug Administration (FDA), spices are "aromatic vegetable substances, in the whole, broken, or ground form, whose significant function in food is seasoning rather than nutrition." The main difference between a herb and a spice is that a spice comes from any part of a plant other than the leaves, while a herb always comes from the leaves

Spices typically come from the dried part of a plant such as buds, flowers (cloves, saffron), bark (cinnamon), root (ginger, Turmeric); fruits/berries (cloves, chili, black pepper); or seeds (cumin) that contain volatile oils or aromatic scents and flavors (Embuscado, 2014). Most of the known herbs and spices originate from Mediterranean countries, the Middle East, or Asia, and many have been used since ancient Egyptian and Roman times (Peter, 2014).

Blue ginger is a ginger-like perennial herb native to Indonesia, India, Sri Lanka, China, Vietnam, Java, and Sumatra. It is known in China by the same names as the lesser Galangal, and also Greeks, Arabs, and Persians do not distinguish it from the latter. The rhizomes are aromatic, refrigerant, mouth freshener, detergent, stomachic, and carminative. They are considered cardiac, liver, and gastric tonic, used in dyspepsia and gastralgia, as a fragrant adjunct to cough and digestive mixtures, useful in rheumatism, catarrhal affections, diabetes insipidus, and as aphrodisiac.

Root and tuber crops grown by small-scale farmers in the region include cassava, sweet potato, and yams, which provide reasonable yields under the recurrent climate change and

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variability and the marginal agroecological conditions of the region. Compared to other staples, root and tuber crops are sources of relatively more significant amounts of energy and nutrition per unit area of land and per unit of time, with some cultivars contributing significant iron, zinc, calcium, and vitamins A and C to diets. They require less-intensive management systems and can be processed into a variety of food, feed, and industrial products ready for household consumption or for sale, Muimba-Kankolongo (2018)

(11)

Preparation and Making of Mangrove Apple Fruit Powder

The raw materials needed for the Proximate and Microbial Analysis of MARC and blue ginger souring mix were gathered and inspected to ensure good quality. Mangrove apples were picked at the slum area. Then, they were rinsed in clean running water, sliced, removed the seeds, and measured using the weighing scale. Then, using a dehydration machine, they were dehydrated for about 6-8 hours.

Preparation and Making of Root Crops Powder

Root crops (gabi, taro, and cassava) was purchased at Bagong Lipunan Trade Center from a trusted supplier. The root crops were washed in clean running water and peeled. Then, they were sliced thinly for easy dehydration. The sliced root crops were arranged on the dehydrator rack and dehydrated for about 4-6 hours.

Preparation and Making of Blue Ginger Powder

Purchasing of blue ginger at Bagong Lipunan Trade Center from a trusted supplier. The blue ginger rhizome was washed in clean running water and peeled. Then, it was sliced thinly to make it easy to dehydrate. The sliced rhizome was arranged on the dehydrator rack and dehydrated for about 4-6 hrs.

Preparation and Formulation of Mangrove Apple-Root Crop and ginger souring Mix.

The tools, materials, ingredients, and workplace have been prepared. The powder from the previous preparation was incorporated and weighed. Then, all the ingredients were formulated and mixed/combined in a blender for the right consistency/mixture. The powder mix was weighed, packed, and labeled.

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(12) Results and Discussions

Sensory Qualities of Mangrove Apple- Root Crop with Blue Ginger Powder Mix

Table 3. Presents the sensory qualities of the Mangrove Apple-Root Crop with Blue

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Ginger Powder Mix, focusing on appearance and texture.: Treatment A (Gabi), Treatment B (Taro), and Treatment C (cassava) were described as "Extremely Appealing." The sensory qualities assessed included appearance and texture, with mean scores provided for each treatment along with their corresponding average deviation (AD). The data further shows that the findings in appearance, Treatment A (gabi) received a mean score of 8.20, Treatment B (taro) received a mean score of 8.40, and Treatment C (Cassava) received a mean score of 8.30, and treatments were described. The mean scores for appearance were relatively high across all treatments, indicating that evaluators generally found the visual presentation of the Mangrove Apple-Root Crop with Blue Ginger Powder Mix to be very appealing. Although there were slight differences in mean scores between treatments, the adjectival description suggests that these differences were minimal. The result implies that the sensory qualities of the three (3) treatments as powder had a similar appeal to the panelists regardless of the three root crops used.

Table 3. Sensory qualities of mangrove apple-root crops and ginger souring mix.

Sensory Qualities	Product B Treatment B					
	Product A Treatment A				Product C Treatment C	
	Mean	AD	Mean	AD	Mean	AD
Appearance	8.20	EA	8.40	EA	8.30	EA
Texture	8.30	EF	8.40	EF	8.20	EF

Legend: **Adjectival Description (AD)**

Score Appearance Texture

8.12 – 9.00	Extremely Appealing (EA)	Extremely Fine (EF)
7.23 – 8.11	Very Much Appealing (VMA)	Very Much Fine (VMF)
6.34 – 7.22	Moderately Appealing (MA)	Moderately Fine (MF)
5.45 – 6.33	Slightly Appealing (SA)	Slightly Fine (SF)

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Sensory Qualities of Mangrove Apple-Root Crop and Blue Ginger souring Mix when Applied to Soup

Table 3.1 presents the sensory qualities of the Mangrove Apple-Root Crop with Blue Ginger Powder Mix when applied to soup across three different treatments: Treatment A (Gabi), Treatment B (Taro), and Treatment C (Cassava). The sensory qualities evaluated

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include appearance, aroma, and taste, with mean scores provided for each treatment along with their corresponding adjectival descriptions (AD) based on a scoring scale. Treatments A, B, and C were described as "Extremely Appealing," with a mean score of ranges from 8.50 – 8.70. The data further shows that in terms of appearance, Treatment A (Gabi) received a mean score of 8.70, Treatment B (Taro) received a mean score of 8.50, and Treatment C (Cassava) received a mean score of 8.90. All treatments were described as "Extremely Appealing." The result means that the evaluators rated the appearance of the soup across all treatments as extremely appealing. Treatment C (Cassava) received the highest mean score, indicating that participants found the appearance of the soup with cassava as the base ingredient to be slightly more appealing compared to the other treatments. This implies that the sensory qualities of the three treatments of powder mix when applied to soup had a similar appeal to the panelists regardless of the three root crops used.

In terms of aroma Treatment, A (Gabi) received a mean score of 8.50, Treatment B (Taro) received a mean score of 8.40, and Treatment C (Cassava) received a mean score of 8.70. Moreover, all treatments are described as "Extremely Pleasant (EP)." Treatment C (Cassava) received the highest mean score, which implies that evaluators perceived the aroma of the soup with cassava as the base ingredient to be slightly more pleasant than the other treatments.

In terms of taste, Treatment A (Gabi) received a mean score of 8.50, Treatment B (Taro) received a mean score of 8.40, and Treatment C (Cassava) received the highest mean score of 8.70. All treatments were described as "Extremely Sour (ES)"; this means that the evaluators rated the taste of the soup across all treatments as extremely sour. Treatment C (Cassava) received the highest mean score, indicating that participants perceived the soup with cassava as the base ingredient to be slightly sourer compared to the other treatments.

Table 3.1. Sensory qualities of mangrove apple-root crops and Blue ginger souring mix when applied to soup.

Sensory Qualities	Treatment A (Gabi)		Treatment B (Taro)		Treatment C (Cassava)	
	Mean	AD	Mean	AD	Mean	AD
Appearance	8.70	EA	8.50	EA	8.90	EA
Aroma	8.50	EP	8.40	EP	8.70	EP
Taste	8.50	ED	8.40	ED	8.70	ED

Legend: **Adjectival Description (AD)**

Score	Appearance	Aroma	Taste
8.12 – 9.00	Extremely Appealing (EA)	Extremely Pleasant (EP)	Extremely Sour (ES)

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Sour (ES)

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7.23 – 8.11 *Very Much Appealing (VMA) Very Much Pleasant (VMP) Very Much Sour (MVS)*
6.34 – 7.22 *Moderately Appealing (MA) Moderately Pleasant (MP) Moderately Sour (MS)*

General Acceptability of Mangrove Appl-Root Crop and Blue Ginger souring Mix

Table 4 revealed the results in the general acceptability of Mangrove Apple-Root Crop with Blue Ginger Powder Mix in terms of sensory qualities as evaluated by a group of consumers using the best treatment of each product. The evaluation was based on various quality attributes such as appearance, texture, and overall acceptability, with scores categorized into qualitative descriptions ranging from liked extremely (LE) to liked slightly (LS). Looking at the mean scores, Treatment B (taro) received the highest mean scores across all quality attributes compared to the other treatments. Specifically, it scored 8.52 for appearance, 8.24 for texture, and 8.38 for general acceptability. These scores fall within the Liked Extremely (LE) and Liked Very Much (LVM) categories, indicating a high level of acceptability among consumers.

Treatment C (Cassava) also performed well, with mean scores close to those of Treatment B, scoring 8.51 for appearance, 8.22 for texture, and 8.36 for general acceptability. Similarly, these scores indicate a high level of acceptability among consumers, falling within the “Liked Extremely (LE)” and “Liked Very Much (LVM)” categories.

Treatment A (Gabi), while still performing well, received slightly lower scores compared to Treatments B and C. It scored 8.16 for appearance, 7.91 for texture, and 8.03 for general acceptability. However, these scores still fall within the LVM category, suggesting that consumers generally liked this treatment. Overall, the findings indicate that all three treatments of mangrove apple-root crops with blue ginger powder mix are well accepted by consumers, with Treatment B (Taro) being the most preferred based on the mean scores for appearance, texture, and general acceptability.

Table 4. General Acceptability of Mangrove Apple-Root Crops and Blue Ginger souring Mix as evaluated by consumers.

Attributes	Treatment B					
	Treatment A (Gabi)		(Taro) Quality		Treatment C (Cassava)	
	Mean	QD	Mean	QD	Mean	QD
Appearance	8.16	LE	8.52	LE	8.51	LE
Texture	7.91	LVM	8.24	LE	8.22	LE

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Overall Acceptability 8.03 LVM 8.38 LE 8.36 LE

Legend: *Qualitative Description (QD)*

Score General Acceptability

8.12 – 9.00 *Liked Extremely (LE)*

7.23 – 8.11 *Liked Very Much (LVM)*

6.34 – 7.22 *Liked Moderately (LM)*

5.45 – 6.33 *Liked Slightly (LS)*

General Acceptability of Mangrove Apple-Root Crops and Blue Ginger souring Mix when Applied in Soup

Table 5. Presents the evaluation of the general acceptability of mangrove apple-root crops with blue ginger powder mix when applied in soup, as assessed by consumers. The evaluation focuses on three different treatments: Gabi (Taro), Taro, and Cassava. Various quality attributes such as appearance, aroma, taste, and overall acceptability were considered, with scores categorized into qualitative descriptions ranging from Liked Extremely (LE) to Liked Slightly (LS).

In terms of appearance, Treatment B (Taro) received the highest mean score of 8.37, falling within the Liked Extremely (LE) category, indicating that consumers highly appreciated the appearance of the soup made with Taro. Treatment C (Cassava) also performed well in this aspect, with a mean score of 8.19, falling within the LE category.

Treatment A (Gabi) received a slightly lower mean score of 7.68, falling within the Liked Very Much (LVM) category, indicating a slightly lower level of appreciation for its appearance compared to the other treatments.

Regarding aroma, Treatment B (Taro) again received the highest mean score of 8.12, indicating that consumers particularly enjoyed the aroma of the soup made with Taro. Treatments A (Gabi) and C (Cassava) received mean scores of 7.66 and 8.02, respectively, falling within the LVM and LE categories, indicating generally positive evaluations of aroma for both treatments.

In terms of taste, Treatment B (Taro) received the highest mean score of 8.43, falling within the LE category, indicating that consumers highly enjoyed the taste of the soup made with Taro. Treatment C (Cassava) also performed well in this aspect, with a mean score of 8.35, falling within the LE category. Treatment A (Gabi) received a slightly lower mean score of 7.51, falling within the LVM category, indicating a slightly lower level of enjoyment of taste compared to the other treatments.

Overall acceptability scores reflect a combination of consumers' evaluations of appearance, aroma, and taste. Treatment B (Taro) received the highest mean score of 8.31, falling within the LE category, indicating that consumers found the soup made with Taro to be highly acceptable overall. Treatment C (Cassava) also performed well in this aspect, with a mean score of 8.19, falling within the LE category. Treatment A (Gabi) received a slightly lower mean score of 7.62, falling within the LVM category, indicating slightly lower overall acceptability compared to the other treatments.

In summary, the findings suggest that soup made with Taro (Treatment B) was the most preferred among consumers, receiving the highest scores across most quality attributes.

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However, soup made with Cassava (Treatment C) also received high scores, indicating strong consumer acceptance. Soup made with Gabi (Treatment A) received slightly lower scores overall but still demonstrated a level of acceptability among consumers.

The results also have the same interpretation in the study of Shaheryar et al. (2023) when they explored taro starch and powder mix in which favorable results suggest that consumers enjoy the taste, aroma, and overall experience of the soup made with the powder mix, particularly when taro powder mix was used. This indicates that the flavor profile and sensory attributes of the soup resonate well with consumers' preferences. Moreover, the use of taro powder mix in the soup formulation demonstrates compatibility with other ingredients, such as mangrove apple-root crops and blue ginger powder mix. The combination creates a harmonious flavor profile that consumers find appealing, indicating that the ingredients complement each other well.

General acceptability of mangrove apple-root crops and Blue ginger souring mix when applied in soup as evaluated by consumers.

Quality Attributes	Treatment A (Gabi)		Treatment B (Taro)		Treatment C (Cassava)	
	Mean	QD	Mean	QD	Mean	QD
Appearance	7.68	LVM	8.37	LE	8.19	LE LVM
Aroma	7.66	LVM	8.12	LE	8.02	
Taste	7.51	LVM	8.43	LE	8.35	LE
Acceptability	7.62	LVM	8.31	LE	8.19	LE

Legend: Qualitative Description (QD)

Score General Acceptability

8.12 – 9.00 Liked Extremely (LE)

7.23 – 8.11 Liked Very Much (LVM)

6.34 – 7.22 Liked Moderately (LM)

(LS)

5.45 – 6.33

Liked Slightly

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pH-Level Testing Mangrove Apple-Root Crops and Blue Ginger souring Mix

Table 6 shows the result of pH level content of mangrove apple-root crop with blue ginger powder mix. The data showed that the mangrove apple-root crop with blue ginger

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powder mixed with a 25g sample obtained a 2.66 pH level, which means that 2.66 is less than 7 ($2.66 < 7$), which means the substance is acidic.

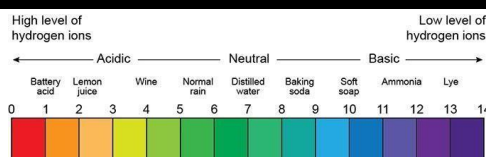
If the pH scale is less than 7, it means the substance is acidic. If the pH level is 7, it means the substance is neutral, and if the pH scale is more than 7, it denotes the substance is a base. The measurement was originally used by the Danish biochemist S.P.L. Sørensen to represent the hydrogen ion concentration, expressed in equivalents per liter, of an aqueous solution: $\text{pH} = -\log[\text{H}^+]$ (in expressions of this kind, enclosure of a chemical symbol within square brackets denotes that the concentration of the symbolized species was the quantity being considered). <https://www.careerpower.in/blog/ph-values-list>.

This data implies that the product's shelf life can last longer since the higher the pH level, the more it prevents bacterial growth. The rancid effect gives the product a much higher quality when stored, and it can also be used as a safer and natural souring agent.

Table 6. pH Level of mangrove apple-root crop and blue ginger souring mix.

Drying Process	Test	Mean
Dehydration	pH (Electronic Method)	pH= 2.66

Legend: (Sajin et al., 2020)



Microbial Analysis of the Best Treatment of Mangrove Apple-Root Crop and Blue Ginger souring Mix

Table 7. shows the result of the laboratory analysis for Microbial in Mangrove apple-root crop with Blue Ginger powder mix conducted by the Negros Prawn Producers Cooperative Analytical and Diagnostic Laboratory, Door No. 1&2., NOLKFI Bldg., 6th Street., Brgy. 7, Bacolod City. The data show that the total coliforms (CFU/G), yeast

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(CFU/G), and Molds (CFU/g) result is Less than ten (<10) and Salmonella spp result Not Detected in 25 g sample. This means that the mangrove apple-root crop with blue ginger powder mix is safe for human consumption.

Table 7. Microbial analysis of the best treatment of mangrove apple-root crop and blue ginger souring mix with BFAD reference for food safety consumption.

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Sample Description	Parameter	Result
Mangrove Apple-Root Crop and Blue Ginger souring	Total	<10 mix
Coliforms		
(1 pack @ 250g	Salmonella spp	Absent in 25
MFD: 05/20/2024)		grams
	Molds Count	<10
	Yeast count	<10

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Proximate Analysis of the Best Treatment of Mangrove Apple-Root Crops and Blue Ginger souring Mix

Table 8. shows the result of the laboratory analysis for Proximate in the process of Mangrove apple-root crop with Blue Ginger powder mix conducted by the Negros Prawn Producers Cooperative Analytical and Diagnostic Laboratory, Door No. 1&2., NOLKFI Bldg., 6th Street., Brgy. 7, Bacolod City. The data shows the result for % Protein is 2.39 followed by % Fat 1.55, % Ash 2.68, % Carbohydrate 26.18, and % moisture with 7.44. The results given in this report are those obtained at the time of examination and refer only to the particular sample submitted.

Table 8. Proximate analysis of the best treatment of mangrove apple-root crop and blue ginger souring mix

Sample Description		Parameter	Result
250g sample per pack Mangrove Apple-Root Crop and Blue Ginger	% Protein	2.39	labeled as:
			% Fat 1.55
	% Ash	2.68	souring mix
	% Carbohydrate		26.18
	% Moisture		7.44

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(13) Conclusion and Recommendation

Conclusion

Semi-trained panelists evaluated powdered seasoning mix formulations, finding variations in appearance and texture. Treatment B was found to be the most visually appealing and palatable. The sensory evaluation of soup revealed variations in appearance, aroma, and taste, with all combinations enhancing the overall sensory experience, with Treatment C excelling in all.

The general acceptability of the product, Consumers' preferences for powdered seasoning mix's appearance and texture were analyzed, with Treatment B (with Taro Root Crops) being more acceptable. When applied in soup's sensory attributes, including appearance, aroma, and taste, were favored in Treatment B (Taro).

The pH level of mangrove apple-root crop with blue ginger powder mix, obtained a 2.66 pH level which means it less than seven (<7) it means the substance was acidic.

The best-performing treatment of the powdered seasoning mix was subjected to microbial analysis and proximate analysis to ensure safety, nutritional quality, and compliance with regulatory standards. It passed the regulatory standard for safety consumption with added nutritional content. Upon receiving the report, it was found that the powder mix was safe for human consumption and contained substantial amounts of nutrients.

Recommendation

- The study on Mangrove apples-root crop and Blue Ginger souring mix suggests researchers should explore different ratios and combinations to achieve desired flavor, nutritional balance, and texture.
- They should also experiment with processing techniques to improve texture and incorporate natural flavor enhancers.
- Innovative packaging solutions should be considered to preserve the mix's freshness and quality.
- Consumer education about the mix's nutritional benefits should be promoted through marketing campaigns.
- Quality control measures should be implemented to ensure consistency and safety.
- Sustainable ingredients should be sourced responsibly, and consumer feedback should be used to refine the souring mix formulation.
- Invest in research and development initiatives to explore new ingredients, processing techniques, and product innovations that align with consumer trends and preferences.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.

Mangrove Apple-root crops and blue ginger souring Mix can be produced and commercialized

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(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

The product of this study could be one of the sources of extension activities conducted by the University, particularly on the main campus.

(16) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

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